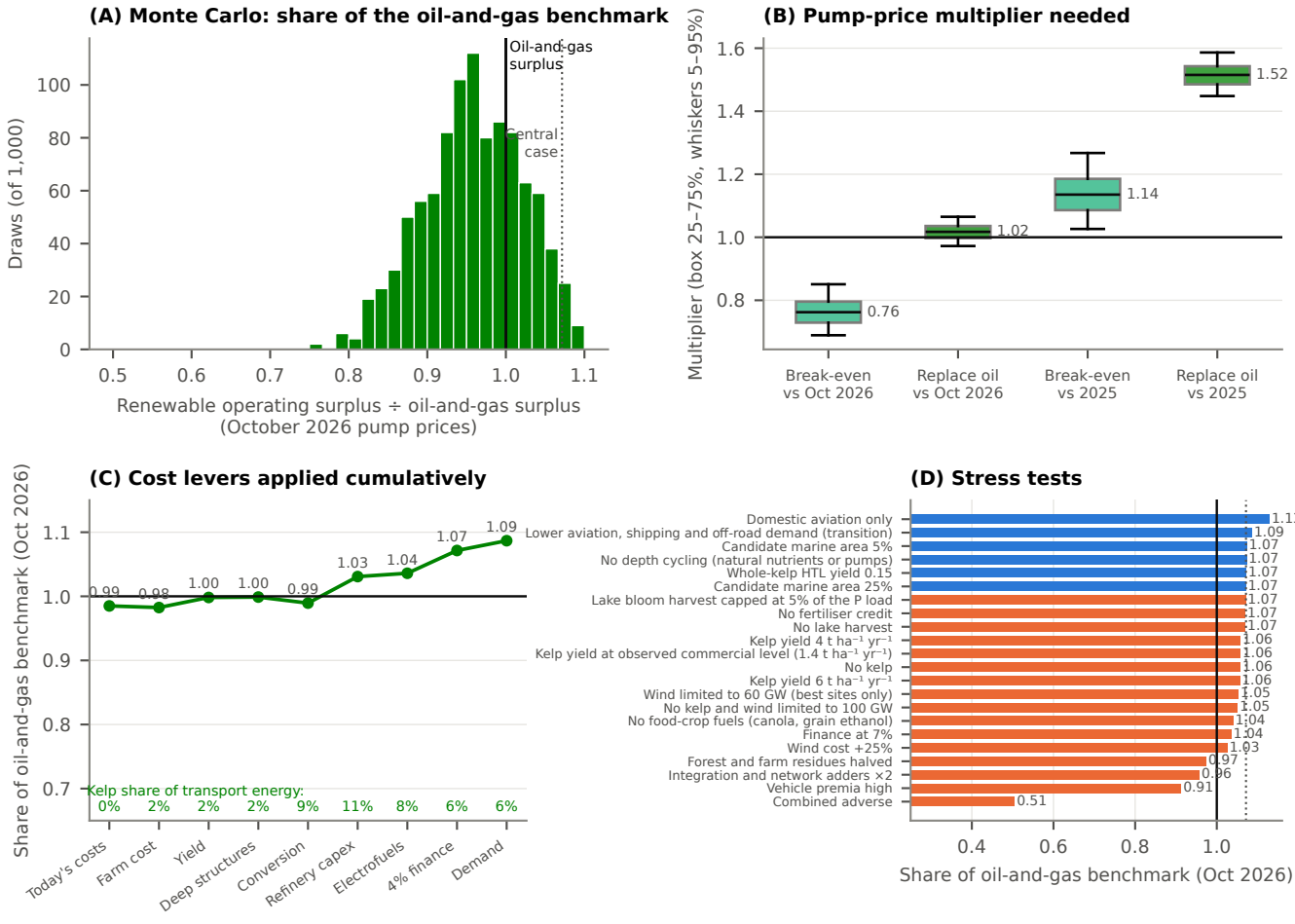
(A) Monte Carlo: share of the oil-and-gas benchmark
Draws (of 1,000)
Oil-and-gas surplus
Central case
Renewable operating surplus ÷ oil-and-gas surplus (October 2026 pump prices)
(B) Pump-price multiplier needed
Multiplier (box 25–75%, whiskers 5–95%)
0.76
1.02
1.14
1.52
Break-even vs Oct 2026
Replace oil vs Oct 2026
Break-even vs 2025
Replace oil vs 2025
(C) Cost levers applied cumulatively
Share of oil-and-gas benchmark (Oct 2026)
0.99
0.98
1.00
1.00
0.99
1.03
1.04
1.07
1.09
Kelp share of transport energy:
0%
2%
2%
2%
9%
11%
8%
6%
6%
Today's costs
Farm cost
Yield
Deep structures
Conversion
Refinery capex
Electrofuels
4% finance
Demand
(D) Stress tests
Domestic aviation only
Lower aviation, shipping and off-road demand (transition)
Candidate marine area 5%
No depth cycling (natural nutrients or pumps)
Whole-kelp HTL yield 0.15
Candidate marine area 25%
Lake bloom harvest capped at 5% of the P load
No fertiliser credit
No lake harvest
Kelp yield 4 t ha⁻¹ yr⁻¹
Kelp yield at observed commercial level (1.4 t ha⁻¹ yr⁻¹)
No kelp
Kelp yield 6 t ha⁻¹ yr⁻¹
Wind limited to 60 GW (best sites only)
No kelp and wind limited to 100 GW
No food-crop fuels (canola, grain ethanol)
Finance at 7%
Wind cost +25%
Forest and farm residues halved
Integration and network adders ×2
Vehicle premia high
Combined adverse
1.13
1.09
1.07
1.07
1.07
1.07
1.07
1.07
1.07
1.06
1.06
1.06
1.06
1.05
1.05
1.04
1.04
1.03
0.97
0.96
0.91
0.51
Share of oil-and-gas benchmark (Oct 2026)